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Sensor actuator with length-adjustable wires and camera module including the same

Abstract

A sensor actuator is provided. The sensor actuator includes a movable body on which an image sensor having an imaging plane is disposed; a fixed body configured to accommodate the movable body; and a driver configured to provide a driving force to move the image sensor, wherein the driver includes a wire portion having a plurality of wires of which lengths change when power is applied to the plurality of wires, wherein each of the plurality of wires is configured to have a first end coupled to the fixed body and a second end coupled to the movable body, and wherein one of the first end and the second end of each of the plurality of wires is connected to the fixed body or the movable body through an elastic portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit under 35 USC § 119(a) of Korean Patent Application No. 10-2022-0002965, filed on Jan. 7, 2022, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference for all purposes.

BACKGROUND

1. Field

(2) The following description relates to a sensor actuator and a camera module including the same.

2. Description of Related Art

(3) Recently, camera modules have been implemented in mobile communication terminals such as, but not limited to, smartphones, tablet personal computers (PCs), and laptops.

(4) Additionally, camera modules may include an actuator which has a focus adjustment function or operation, and an optical image stabilization (OIS) function or operation to generate a high-resolution image.

(5) For example, a focus may be adjusted by moving a lens module in the optical axis (Z-axis) direction, or shaking may be corrected by moving the lens module in a direction perpendicular to the optical axis (Z-axis).

(6) However, recently, as the performance of camera modules has improved, the weight of the lens module has increased, and due to weight of a driver that moves the lens module, it may be difficult to precisely control a driving force necessary for image stabilization.

SUMMARY

(7) This Summary is provided to introduce a selection of concepts in a simplified form that is further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(8) In a general aspect, a sensor actuator includes a movable body on which an image sensor having an imaging plane is disposed; a fixed body, configured to accommodate the movable body; and a driver, configured to provide a driving force to move the image sensor, wherein the driver comprises a plurality of wires which have lengths that vary when power is applied, wherein each of the plurality of wires is configured to have a first end coupled to the fixed body and a second end coupled to the movable body, and wherein one of the first end and the second end of each of the plurality of wires is connected to the fixed body or the movable body through an elastic portion.

(9) The fixed body may include a first support plate, and the movable body comprises a second support plate, and the elastic portion may be disposed on one of the first support plate and the second support plate.

(10) The wire portion may include a first wire portion, a second wire portion, a third wire portion, and a fourth wire portion, wherein each of the first to fourth wire portions may include two wires, and wherein each of the first to fourth wire portions may be configured to have the first end fixed to the fixed body and the second end fixed to the movable body.

(11) Each of the first wire portion and the second wire portion may be configured to provide a driving force to move the movable body in a first direction parallel to the imaging plane, and a moving direction of the movable body based on an operation of the first wire portion and a moving direction of the movable body based on an operation of the second wire portion are opposite to each other.
(12) A length of one of the first wire portion and the second wire portion may be reduced when power is applied to the one of the first wire portion and the second wire portion, and wherein the elastic portion connected to wire portions other than the one of the first wire portion and the second wire portion is elastically deformed.
(13) Each of the third wire portion and the fourth wire portion may be configured to provide a driving force to move the movable body in a second direction parallel to the imaging plane, and a moving direction of the movable body based on an operation of the third wire portion and a moving direction of the movable body based on an operation of the fourth wire portion may be opposite to each other.
(14) A length of one of the third wire portion and the fourth wire portion may be reduced when power is applied to the one of the third wire portion and the fourth wire portion, and wherein the elastic portion connected to wire portions other than the one of the third wire portion and the fourth wire portion may be elastically deformed.
(15) The movable body may be configured to rotate by at least two wires which are configured to generate driving forces in opposite directions.
(16) Each of the first wire portion and the second wire portion may be configured to provide a driving force to move the movable body in a first direction parallel to the imaging plane, wherein the two wires of the first wire portion and the two wires of the second wire portion may be spaced apart from each other in a second direction parallel to the imaging plane, and wherein the first direction and the second direction are perpendicular to each other.
(17) Each of the third wire portion and the fourth wire portion may be configured to provide a driving force to move the movable body in the second direction, respectively, and wherein the two wires of the third wire portion and the two wires of the fourth wire portion may be spaced apart from each other in the first direction.
(18) A ball member, configured to support movement of the movable body, may be disposed between the movable body and the fixed body.
(19) The sensor actuator may include a support substrate configured to support the movable body such that the movable body moves, wherein the support substrate comprises a deformable portion that is elastically deformed as the movable body moves.
(20) The support substrate may further include a movable portion on which the movable body is disposed, and a fixed portion coupled to the fixed body, and the elastic portion elastically connects the movable portion to the fixed portion.
(21) The sensor actuator may further include a position sensing portion configured to sense a position of the image sensor, and comprising a sensing coil disposed on one of the movable body and the fixed body, and a sensing yoke portion disposed on the other of the movable body and the fixed body, wherein the sensing yoke portion comprises a plurality of sensing yokes spaced apart from each other in a direction parallel to the imaging plane, and wherein each sensing yoke is configured to have a width that changes in a moving direction of the image sensor.
(22) The plurality of sensing yokes may include a first sensing yoke and a second sensing yoke, wherein each of the first sensing yoke and the second sensing yoke opposes the sensing coil in a direction perpendicular to the imaging plane, and wherein each of the first sensing yoke and the second sensing yoke has a width that increases and decreases in a moving direction of the image sensor, and positions of the first sensing yoke and the second sensing yoke in which widths increase or decrease have different shapes.
(23) In a general aspect, a camera module includes a lens module including at least one lens; a

housing configured to accommodate the lens module; a first driver configured to move the lens module in an optical axis direction; a fixed body coupled to the housing; a movable body accommodated in the fixed body and comprising an image sensor disposed therein; a second driver configured to provide a driving force to move the image sensor in a first direction and a second direction perpendicular to the optical axis direction, wherein the second driver comprises a plurality of wires which have lengths that vary when power is applied, wherein each of the plurality of wires is configured to have a first end coupled to the fixed body and a second end coupled to the movable body, and wherein one of the first end and the second end of each wire of the plurality of wires is connected to the fixed body or the movable body through an elastic portion.
(24) In a general aspect, an apparatus includes a camera module, including: a movable body on which an image sensor is disposed; a fixed body configured to accommodate the movable body; a driver, configured to provide a driving force to move the image sensor, the driver including: wire portions comprising first ends coupled to the fixed body and second ends coupled to the movable body, and configured to provide a driving force to move the moveable body in first and second directions parallel to an imaging plane of the image sensor, and configured to rotate the movable body about an optical axis; wherein each of the wire portions has a length that varies when power is applied.
(25) Each of the wire portions is a shape memory alloy.
(26) The first direction may include directions that are opposite to each other, the second direction may include directions that are opposite to each other, and the first direction is perpendicular to the second direction.
(27) Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a cross-sectional diagram illustrating an example sensor actuator, in accordance with one or more embodiments.
(2) FIG. **2**, FIG. **3**, and FIG. **4** are cross-sectional diagrams illustrating an example sensor actuator, in accordance with one or more embodiments.
(3) FIG. **5** is a perspective diagram illustrating an example in which a driver is connected to a fixed body and a movable body, in accordance with one or more embodiments.
(4) FIG. **6** is a plan diagram illustrating a second support plate, in accordance with one or more embodiments.
(5) FIG. **7** is a perspective diagram illustrating an example in which a driver is connected to a first support plate and a second support plate, in accordance with one or more embodiments.
(6) FIG. **8** is a perspective diagram illustrating an example in which a portion of a driver is fixed to a first support plate and a second support plate, in accordance with one or more embodiments.
(7) FIG. **9**A, FIG. **9**B, FIG. **10**A, FIG. **10**B, FIG. **11**A, and FIG. **11**B are diagrams illustrating a driving force direction of a driver, in accordance with one or more embodiments.
(8) FIG. **12**A, FIG. **12**B, FIG. **13**A, FIG. **13**B, FIG. **14**A, and FIG. **14**B are diagrams illustrating a driving force direction of a driver, in accordance with one or more embodiments.
(9) FIG. **15** is a diagram illustrating a configuration of a position sensing portion, in accordance with one or more embodiments.
(10) FIG. **16** is a diagram illustrating a sensing yoke portion and a sensing coil of a position sensing portion, in accordance with one or more embodiments.
(11) FIGS. **17**A and **17**B are diagrams illustrating changes in a positional relationship between a first sensing yoke portion and a first sensing coil, in accordance with one or more embodiments.

(12) FIG. **18** is a graph illustrating inductance of a first sensing coil according to movement of an image sensor in one direction, in accordance with one or more embodiments.
(13) FIG. **19**A is a graph illustrating a plurality of levels of inductance of a first sensing coil corresponding to a first sensing yoke and a second sensing yoke of a sensor actuator, respectively, in accordance with one or more embodiments.
(14) FIG. **19**B is a graph illustrating arctangent processing values of a plurality of levels of inductance illustrated in FIG. **19**A.
(15) FIG. **20** is a cross-sectional diagram illustrating an example camera module, in accordance with one or more embodiments.
(16) FIG. **21** is a cross-sectional diagram illustrating an example camera module, in accordance with one or more embodiments.
(17) Throughout the drawings and the detailed description, the same reference numerals may refer to the same, or like, elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

(18) The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent after an understanding of the disclosure of this application. For example, the sequences of operations described herein are merely examples, and are not limited to those set forth herein, but may be changed as will be apparent after an understanding of the disclosure of this application, with the exception of operations necessarily occurring in a certain order. Also, descriptions of features that are known, after an understanding of the disclosure of this application, may be omitted for increased clarity and conciseness, noting that omissions of features and their descriptions are also not intended to be admissions of their general knowledge.
(19) The features described herein may be embodied in different forms, and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided merely to illustrate some of the many possible ways of implementing the methods, apparatuses, and/or systems described herein that will be apparent after an understanding of the disclosure of this application.
(20) Although terms such as “first,” “second,” and “third” may be used herein to describe various members, components, regions, layers, or sections, these members, components, regions, layers, or sections are not to be limited by these terms. Rather, these terms are only used to distinguish one member, component, region, layer, or section from another member, component, region, layer, or section. Thus, a first member, component, region, layer, or section referred to in examples described herein may also be referred to as a second member, component, region, layer, or section without departing from the teachings of the examples.
(21) Throughout the specification, when an element, such as a layer, region, or substrate, is described as being “on,” “connected to,” or “coupled to” another element, it may be directly “on,” “connected to,” or “coupled to” the other element, or there may be one or more other elements intervening therebetween. In contrast, when an element is described as being “directly on,” “directly connected to,” or “directly coupled to” another element, there can be no other elements intervening therebetween. Likewise, expressions, for example, “between” and “immediately between” and “adjacent to” and “immediately adjacent to” may also be construed as described in the foregoing.
(22) The terminology used herein is for the purpose of describing particular examples only, and is not to be used to limit the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. As used herein, the term “and/or” includes any one and any combination of any two or more of the

associated listed items. As used herein, the terms “include,” “comprise,” and “have” specify the presence of stated features, numbers, operations, elements, components, and/or combinations thereof, but do not preclude the presence or addition of one or more other features, numbers, operations, elements, components, and/or combinations thereof. The use of the term “may” herein with respect to an example or embodiment (for example, as to what an example or embodiment may include or implement) means that at least one example or embodiment exists where such a feature is included or implemented, while all examples are not limited thereto.
(23) Unless otherwise defined, all terms, including technical and scientific terms, used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure pertains consistent with and after an understanding of the present disclosure. Terms, such as those defined in commonly used dictionaries, are to be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and are not to be interpreted in an idealized or overly formal sense unless expressly so defined herein.
(24) A sensor actuator **100**, in accordance with one or more embodiments, may be one of multiple components of a camera module. Additionally, the camera module may be mounted on a portable electronic device. A portable electronic device may be implemented as a portable electronic device such as a mobile communication terminal, a smartphone, or a tablet PC.
(25) FIG. **1** is a cross-sectional diagram illustrating a sensor actuator, in accordance with one or more embodiments.
(26) Referring to FIG. **1**, the sensor actuator **100**, in accordance with one or more embodiments, may include a movable body **110**, a fixed body **130**, and a driver **120**.
(27) The movable body **110** may include a sensor substrate **112**. The movable body **110** may be disposed to move relative to the fixed body **130**.
(28) An image sensor **111** may be disposed on a first surface of the sensor substrate **112**, and a reinforcing plate **113**, that reinforces the rigidity of the sensor substrate **112**, may be coupled to a second surface of the sensor substrate **112**. Since the movable body **110** moves with the reinforcing plate **113**, the sensor substrate **112** and the fixed body **130** may be prevented from being in contact with each other.
(29) The sensor substrate **112** may be connected to the connector substrate **190** to transmit a signal of the image sensor **111** to an external entity. For example, the sensor substrate **112** and the connector substrate **190** may be electrically connected to each other by the connection portion **191**. The connection portion **191** may include a plurality of bridges flexibly bent according to movement of the image sensor **111**.
(30) A signal of the image sensor **111** may be transmitted to the other electronic components through the sensor substrate **112**, the connection portion **191**, and the connector substrate **190**.
(31) In an example embodiment, the sensor substrate **112** may have an accommodation space to accommodate the image sensor **111**. In an example, the accommodation space may have a form of a groove or a hole. The image sensor **111** may be disposed in the accommodation space and may be electrically connected to the sensor substrate **112**.
(32) The fixed body **130** of the actuator may include a housing **131**. The housing **131** may have an internal space to accommodate the movable body **110**.
(33) The driver **120** may move the movable body **110**.
(34) The movable body **110** may move in a direction perpendicular to the direction in which the imaging plane **111***a* of the image sensor **111** is oriented based on an operation of the driver **120**. In an example embodiment, the driver **120** may move the image sensor **111** to compensate for shaking that occurs when the camera modules **10** and **20** (FIGS. **20** and **21**) on which the image sensor **111** is mounted perform imaging.
(35) The driver **120** may move the movable body **110** including the image sensor **111** in a first direction (X-direction) and a second direction (Y-direction) perpendicular to the optical axis (Z-axis). The first direction (X-direction) and the second direction (Y-direction) may intersect each

other. For example, the driver **120** may allow the movable body **110** to move in the first direction (X-direction) and/or the second direction (Y-direction) perpendicular to the optical axis (Z-axis), thereby correcting shaking. Additionally, the driver **120** may rotate the movable body **110** using the optical axis (Z-axis) as a rotation axis.
(36) The driver **120** may include a wire portion of which a length may change as power is applied. The wire portion may include a plurality of wires, and each of the plurality of wires may be a shape memory alloy.
(37) In example embodiments, the direction in which the imaging plane **111***a* of the image sensor **111** is oriented may be referred to as the optical axis (Z-axis) direction. That is, the movable body **110** may move in a direction perpendicular to the optical axis (Z-axis) with respect to the fixed body **130**.
(38) In the drawings, the configuration in which the movable body **110** may move in a direction parallel to the imaging plane **111***a* of the image sensor **111** may indicate that the movable body **110** may move in a direction perpendicular to the optical axis (Z-axis).
(39) The configuration in which the movable body **110** may move in the first direction (X-direction) may indicate that the movable body **110** may move in a direction perpendicular to the optical axis (Z-axis).
(40) Additionally, the first direction (X-direction) and the second direction (Y-direction) may be examples of two directions perpendicular to the optical axis (Z-axis) and intersecting each other, and in example embodiments, the first direction (X-direction) and the second direction (Y-direction) may be understood as two directions perpendicular to the optical axis (Z-axis) and intersecting each other.
(41) In an example embodiment, the ball member B may be disposed between the fixed body **130** and the movable body **110**.
(42) The first guide groove **132** and the second guide groove **114**, that accommodate at least a portion of the ball member B, may be provided in the fixed body **130** and the movable body **110**, respectively. For example, the first guide groove **132** and the second guide groove **114** may be formed on a surface on which the fixed body **130** and the movable body **110** oppose each other in the optical axis (Z-axis) direction.
(43) The ball member B may be disposed between the first guide groove **132** of the fixed body **130** and the second guide groove **114** of the movable body **110**. Accordingly, when the movable body **110** moves within the fixed body **130**, the movable body **110** may be guided by the ball member B such that the movable body **110** may move smoothly.
(44) Both the first guide groove **132** and the second guide groove **114** may have a shape that does not limit a rolling direction of the ball member B. For example, the first guide groove **132** and the second guide groove **114** may have a polygonal shape or a circular shape having a size larger than the diameter of the ball member B.
(45) In a non-limiting example, the ball member B may include at least three balls, and each of the first guide groove **132** and the second guide groove **114** may include guide grooves of which the number may correspond to the number of balls included in the ball member B.
(46) The ball member B, the first guide groove **132** and the second guide groove **114** may guide the movable body **110** such that the movable body **110** may translate and/or rotate on the X-Y plane
(47) Referring to FIG. **1**, the ball member B, and the second guide groove **114** may be disposed on the movable body **110**, and the first guide groove **132** may be disposed on the fixed body **130**.
(48) FIGS. **2** to **4** are cross-sectional diagrams illustrating a sensor actuator, in accordance with one or more embodiments.
(49) First, referring to FIG. **2**, in an example, the ball member B, the first guide groove **132** and the second guide groove **114** may be disposed below the movable body **110**.
(50) Additionally, an infrared cut-off filter **140** may be disposed in a position spaced apart from the sensor substrate **112**. The infrared cut-off filter **140** may be disposed with a spacing from the sensor

substrate **112** by the spacer **141**. The spacers **141** may be continuously disposed along a periphery of the imaging plane **111***a*. The imaging plane **111***a* may be protected from the outside by the infrared cut-off filter **140** and the spacer **141**.
(51) Referring to FIG. **3**, the movable body **110** may be coupled to the support substrate **150**. For example, the support substrate **150** may support the sensor substrate **112** to move.
(52) The support substrate **150** may include a movable portion **151** on which the sensor substrate **112** is seated and a fixed portion **152** fixed to the fixed body **130**. Additionally, the support substrate **150** may include a deformable portion **153** that elastically deforms as the movable body **110** moves.
(53) The deformable portion **153** may elastically connect the movable portion **151** to the fixed portion **152**.
(54) The movable body **110** may move together with the movable portion **151** of the support substrate **150**, and as the movable body **110** moves, the deformable portion **153** may be flexibly bent.
(55) Referring to FIG. **4**, the ball member B may be disposed between the movable body **110** and the support substrate **150**.
(56) At least one of the movable body **110** and the support substrate **150** may include a guide groove **114** to accommodate at least a portion of the ball member B. For example, the guide groove **114** may be formed on one surface of the movable body **110** opposing the support substrate **150** in the optical axis (Z-axis) direction.
(57) In an example, the guide groove **114** may be formed in the movable body **110** in FIG. **4**. However, this is merely an example, and the guide groove may also be formed in the support substrate **150**.
(58) FIG. **5** is a perspective diagram illustrating an example in which a driver is connected to a fixed body and a movable body, in accordance with one or more embodiments. FIG. **6** is a plan diagram illustrating a second support plate, in accordance with one or more embodiments.
(59) FIG. **7** is a perspective diagram illustrating an example in which a driver is connected to a first support plate and a second support plate, in accordance with one or more embodiments. FIG. **8** is a perspective diagram illustrating an example in which a portion of a driver is fixed to a first support plate and a second support plate, in accordance with one or more embodiments.
(60) FIGS. **9**A to **11**B are diagrams illustrating a driving force direction of a driver, in accordance with one or more embodiments.
(61) Referring to FIGS. **5** to **8**, the fixed body **130** may include a first fixed plate **131***a*, a second fixed plate **131***b*, and a first support plate **131***c*.
(62) The first support plate **131***c* may be disposed between the first fixed plate **131***a* and the second fixed plate **131***b*, and the first fixed plate **131***a*, the second fixed plate **131***b* and the first support plate **131***c* may be coupled to the housing **131**.
(63) In an example, the first fixed plate **131***a* and the second fixed plate **131***b* may be optional components, and if desired, the first support plate **131***c* may be directly coupled to the housing **131** of the sensor actuator **100**.
(64) The movable body **110** may include a first sensor substrate **112***a*, a second sensor substrate **112***b*, and a second support plate **112***c*.
(65) A second support plate **112***c* may be disposed between the first sensor substrate **112***a* and the second sensor substrate **112***b*, and the image sensor **111** may be disposed on the first sensor substrate **112***a* or the second sensor substrate **112***b*.
(66) One of the first sensor substrate **112***a* and the second sensor substrate **112***b* may be an optional component, and if desired, the movable body **110** may only include the first sensor substrate **112***a* and the second support plate **112***c*, or may only include the second sensor substrate **112***b* and the second support plate **112***c*.
(67) The driver **120** may include a plurality of wires, and the lengths of the plurality of wires may

change as power is applied to each of the plurality of wires. Each of the plurality of wires may have a first end coupled to the fixed body **130**, and a second end coupled to the movable body **110**.
(68) For example, each of the plurality of wires may have a first end coupled to the first support plate **131***c*, and a second end coupled to the second support plate **112***c*.
(69) A substrate **131***d* (FIG. **8**) that supplies power to the plurality of wires may be coupled to the first support plate **131***c* and the second support plate **112***c*. For example, referring to FIG. **8**, a substrate **131***d* may be coupled to the first support plate **131***c*, and one end of the plurality of wires may be fixed to the substrate **131***d*. Although not illustrated in FIG. **8**, a substrate that supplies power to the plurality of wires may also be coupled to the second support plate **112***c*.
(70) Referring to FIG. **6**, the second support plate **112***c* may include a body portion **112***d*, a support portion **112***e*, and an elastic portion **112***f*. In an example, the body portion **112***d* may have a plate shape surrounding the image sensor **111**, and the support portion **112***e* may have a shape protruding from the body portion **112***d*. A plurality of wires may be fixed to the support portion **112***e*. The body portion **112***d* and the support portion **112***e* may be connected to each other by the elastic portion **112***f*. The elastic portion **112***f* may have a shape which may be bent according to changes in lengths of the plurality of wires.
(71) One of a first end and a second end of each wire may be connected to the movable body **110** through the elastic portion **112***f*.
(72) In the example embodiment, the elastic portion **112***f* may be a component of the movable body **110**, but this is merely an example, and the elastic portion **112***f* may be formed on the first support plate **131***c* and may be provided as a component of the fixed body **130**.
(73) Referring to FIGS. **9**A to **11**B, the driver **120** may include a first wire portion **121**, a second wire portion **122**, a third wire portion **123**, and a fourth wire portion **124**.
(74) Each of the first wire portion **121** to the fourth wire portion **124** may include a plurality of wires of which the lengths thereof may change when power is applied. For example, each of the first wire portion **121** to the fourth wire portion **124** may include two wires. In a non-limited example, the wire may be a shape memory alloy.
(75) Each of the first wire portion **121** to the fourth wire portion **124** may have a first end that is fixed to the fixed body **130**, and a second end that is fixed to the movable body **110**. Additionally, the second end of each wire portion may be connected to the movable body **110** by the elastic portion **112***f*.
(76) Accordingly, the movable body **110** may move relative to the fixed body **130** based on a change in the length of each wire portion.
(77) The first wire portion **121** and the second wire portion **122** may provide driving forces to move the movable body **110** in a first direction (X-direction) parallel to the imaging plane **111***a*. In an example, the moving direction of the movable body **110** by the first wire portion **121** and the moving direction of the movable body **110** by the second wire portion **122** may be opposite to each other. For example, the movable body **110** may move in the +X-direction based on an operation by the first wire portion **121**, and the movable body **110** may move in the −X-direction based on an operation by the second wire portion **122**.
(78) The third wire portion **123** and the fourth wire portion **124** may provide driving forces to move the movable body **110** in a second direction (Y-direction) parallel to the imaging plane **111***a*. In an example, the moving direction of the movable body **110** based on an operation by the third wire portion **123** and the moving direction of the movable body **110** based on an operation by the fourth wire portion **124** may be opposite to each other. For example, the movable body **110** may move in the +Y-direction based on an operation by the third wire portion **123**, and the movable body **110** may move in the −Y-direction based on an operation by the fourth wire portion **124**.
(79) In an example, each of the first wire portion **121** to the fourth wire portion **124** may include two wires. A first end of each wire may be connected to the fixed body **130**, and a second end of each wire may be connected to the movable body **110**.

(80) The two wires of the first wire portion **121** may be disposed to be spaced apart from each other in the second direction (Y-direction). Additionally, when viewed in the optical axis (Z-axis) direction, the two wires of the first wire portion **121** may have a shape having a length in the first direction (X-direction). For example, the two wires of the first wire portion **121** may extend in the +X-direction from a first end toward a second end, and the second ends of the two wires of the first wire portion **121** may be connected to the movable body through the elastic portion **112***f*.
(81) The two wires of the second wire portion **122** may be disposed to be spaced apart from each other in the second direction (Y-direction). Additionally, when viewed in the optical axis (Z-axis) direction, the two wires of the second wire portion **122** may have a shape having a length in the first direction (X-direction). For example, the two wires of the second wire portion **122** may extend in the −X-direction from a first end toward the second end, and the second ends of the two wires of the second wire portion **122** may be connected to the movable body **110** through the elastic portion **112***f*.
(82) Accordingly, the first wire portion **121** and the second wire portion **122** may be disposed to intersect each other in an X shape when viewed in the second direction (Y-direction).
(83) The two wires of the third wire portion **123** may be disposed to be spaced apart in the first direction (X-direction). Additionally, when viewed in the optical axis (Z-axis) direction, the two wires of the third wire portion **123** may have a shape having a length in the second direction (Y-direction). For example, the two wires of the third wire portion **123** may extend in the +Y-direction from the first end to the second end, and the second ends of the two wires of the third wire portion **123** may be connected to the movable body **110** through the elastic portion **112***f*.
(84) The two wires of the fourth wire portion **124** may be disposed to be spaced apart from each other in the first direction (X-direction). Additionally, when viewed in the optical axis (Z-axis) direction, the two wires of the fourth wire portion **124** may have a shape having a length in the second direction (Y-direction). For example, the two wires of the fourth wire portion **124** may extend in the −Y-direction from the first end to the second end, and the second ends of the two wires of the fourth wire portion **124** may be connected to the movable body **110** through the elastic portion **112***f*.
(85) Accordingly, the third wire portion **123** and the fourth wire portion **124** may be disposed to intersect each other in an X shape when viewed in the first direction (X-direction).
(86) Referring to FIG. **9**A, when power is applied to the first wire portion **121**, a length of the first wire portion **121** may be reduced. Accordingly, the movable body **110** may be pulled in the direction of the arrow based on a change in length of the first wire portion **121**, and accordingly, the movable body **110** may move in the +X-direction.
(87) In this example, the elastic portion **112***f* connected to each of the second ends of the second wire portion **122** to the fourth wire portion **124** may be elastically deformed.
(88) Referring to FIG. **9**B, when power is applied to the second wire portion **122**, a length of the second wire portion **122** may be reduced. Accordingly, the movable body **110** may be pulled in the direction of the arrow based on a change in length of the second wire portion **122**, and accordingly, the movable body **110** may move in the −X-direction.
(89) In this example, the elastic portion **112***f* connected to each of the second ends of the first wire portion **121**, the third wire portion **123**, and the fourth wire portion **124** may be elastically deformed.
(90) Referring to FIG. **10**A, when power is applied to the third wire portion **123**, a length of the third wire portion **123** may be reduced. Accordingly, the movable body **110** may be pulled in the direction of the arrow based on a change in length of the third wire portion **123**, and accordingly, the movable body **110** may move in the +Y-direction.
(91) In this example, the elastic portion **112***f* connected to each of the second ends of the first wire portion **121**, the second wire portion **122**, and the fourth wire portion **124** may be elastically deformed.

(92) Referring to FIG. **10**B, when power is applied to the fourth wire portion **124**, a length of the fourth wire portion **124** may be reduced. Accordingly, the movable body **110** may be pulled in the direction of the arrow based on a change in length of the fourth wire portion **124**, and accordingly, the movable body **110** may move in the −Y-direction.
(93) In this example, the elastic portion **112***f* connected to each of the second ends of the first wire portion **121** to the third wire portion **123** may be elastically deformed.
(94) The movable body **110** may rotate about the optical axis (Z-axis) based on at least two wires that create driving forces in opposite directions.
(95) Referring to FIGS. **11**A and **11**B, by applying power to one wire (four wires) of the two wires included in each wire portion, the movable body **110** may rotate in a clockwise direction or a counterclockwise direction.
(96) FIGS. **11**A and **11**B illustrate an example of rotating the movable body **110** by four wires. However, this is merely an example, and the movable body **110** may rotate by at least two wires that create driving forces in opposite directions.
(97) FIGS. **12**A to **14**B are diagrams illustrating a driving force direction of a driver, in accordance with one or more embodiments.
(98) Referring to FIGS. **12**A to **14**B, the driver **120** may include a first wire portion **121**, a second wire portion **122**, a third wire portion **123**, a fourth wire portion **124**, a fifth wire portion **125** and a sixth wire portion **126**.
(99) Each of the first wire portion **121** to the sixth wire portion **126** may include a wire of which a length changes when power is applied thereto. In an example, the wire may be a shape memory alloy.
(100) A first end of the first wire portion **121** to the sixth wire portion **126** may be fixed to the fixed body **130**, and a second end of the first wire portion **121** to the sixth wire portion **126** may be fixed to the movable body **110**. Additionally, the second end of each wire portion may be connected to the movable body **110** by the elastic portion **112***f*.
(101) Accordingly, the movable body **110** may move relative to the fixed body **130** based on a change in the length of each wire portion.
(102) The first wire portion **121** and the second wire portion **122** may provide a driving force to move the movable body **110** in a first direction (X-direction) parallel to the imaging plane **111***a*. In an example, the moving direction of the movable body **110** based on the first wire portion **121** and the moving direction of the movable body **110** based on the second wire portion **122** may be opposite to each other. For example, the movable body **110** may move in the +X-direction based on the first wire portion **121**, and the movable body **110** may move in the −X-direction by the second wire portion **122**.
(103) The third wire portion **123** and the fourth wire portion **124** may provide a driving force to move the movable body **110** in the second direction (Y-direction) parallel to the imaging plane **111***a*. In an example, the moving direction of the movable body **110** based on the third wire portion **123** and the moving direction of the movable body **110** based on the fourth wire portion **124** may be opposite to each other. For example, the movable body **110** may move in the +Y-direction based on the third wire portion **123**, and the movable body **110** may move in the −Y-direction by the fourth wire portion **124**.
(104) The first wire portion **121** may have a shape having a length in the first direction (X-direction) when viewed in the optical axis (Z-axis) direction. For example, the first wire portion **121** may extend in the +X-direction from a first end toward the second end, and the second end of the first wire portion **121** may be connected to the movable body **110** through the elastic portion **112***f*.
(105) The second wire portion **122** may have a shape having a length in the first direction (X-direction) when viewed in the optical axis (Z-axis) direction. For example, the second wire portion **122** may extend from one end toward the other end in the −X-direction, and the other end of the

second wire portion **122** may be connected to the movable body **110** through the elastic portion **112***f*.
(106) Accordingly, the first wire portion **121** and the second wire portion **122** may be disposed to intersect each other in an X shape when viewed in the second direction (Y-direction).
(107) The third wire portion **123** may have a shape having a length in the second direction (Y-direction) when viewed in the optical axis (Z-axis) direction. For example, the third wire portion **123** may extend from a first end toward a second end in the +Y-direction, and the second end of the third wire portion **123** may be connected to the movable body **110** through the elastic portion **112***f*.
(108) The fourth wire portion **124** may have a shape having a length in the second direction (Y-direction) when viewed in the optical axis (Z-axis) direction. For example, the fourth wire portion **124** may extend from a first end toward a second end in the −Y-direction, and the second end of the fourth wire portion **124** may be connected to the movable body **110** through the elastic portion **112***f*.
(109) Accordingly, the third wire portion **123** and the fourth wire portion **124** may be disposed to intersect each other in an X shape when viewed in the first direction (X-direction).
(110) The first wire portion **121** to the fourth wire portion **124** may be disposed to intersect the center of the imaging plane **111***a*. Accordingly, the first wire portion **121** to the fourth wire portion **124** may be disposed in the space on the lower side of the image sensor **111**, preferably.
(111) Referring to FIG. **12**A, when power is applied to the first wire portion **121**, a length of the first wire portion **121** may be reduced. Accordingly, the movable body **110** may be pulled in the direction of the arrow based on a change in length of the first wire portion **121**, and accordingly, the movable body **110** may move in the +X-direction.
(112) In this example, the elastic portion **112***f* connected to each of the second ends of the second wire portion **122** to the sixth wire portion **126** may be elastically deformed.
(113) Referring to FIG. **12**B, when power is applied to the second wire portion **122**, a length of the second wire portion **122** may be reduced. Accordingly, the movable body **110** may be pulled in the direction of the arrow based on a change in length of the second wire portion **122**, and accordingly, the movable body **110** may move in the −X-direction.
(114) In this example, the elastic portion **112***f* connected to each of the second ends of the first wire portion **121**, the third wire portion **123** to the sixth wire portion **126** may be elastically deformed.
(115) Referring to FIG. **13**A, when power is applied to the third wire portion **123**, a length of the third wire portion **123** may be reduced. Accordingly, the movable body **110** may be pulled in the direction of the arrow based on a change in the length of the third wire portion **123**, and accordingly, the movable body **110** may move in the +Y-direction.
(116) In this example, the elastic portion **112***f* connected to each of the second ends of the first wire portion **121**, the second wire portion **122**, the fourth wire portion **124** to the sixth wire portion **126** may be elastically deformed.
(117) Referring to FIG. **13**B, when power is applied to the fourth wire portion **124**, a length of the fourth wire portion **124** may be reduced. Accordingly, the movable body **110** may be pulled in the direction of the arrow based on a change in length of the fourth wire portion **124**, and accordingly, the movable body **110** may move in the −Y-direction.
(118) In this example, the elastic portion **112***f* connected to each of the second ends of the first wire portion **121** to the third wire portion **123**, the fifth wire portion **125**, and the sixth wire portion **126** may be elastically deformed.
(119) Referring to FIGS. **14**A and **14**B, the movable body **110** may rotate based on the fifth wire portion **125** and the sixth wire portion **126**. For example, the movable body **110** may rotate in a counterclockwise direction based on the fifth wire portion **125**, and the movable body **110** may rotate in a clockwise direction based on the sixth wire portion **126**.
(120) The fifth wire portion **125** and the sixth wire portion **126** may have a shape having a length in the first direction (X-direction) when viewed in the optical axis (Z-axis) direction. For example, the

fifth wire portion **125** and the sixth wire portion **126** may extend in the +X-direction from a first end to a second end, and the second ends of the fifth wire portion **125** and the sixth wire portion **126** may be connected to the movable body **110** through the elastic portion **112***f*. Alternatively, the fifth wire portion **125** and the sixth wire portion **126** may extend in the −X-direction from a first end to the second end.
(121) The fifth wire portion **125** and the sixth wire portion **126** may be spaced apart from each other in a direction parallel to the imaging plane **111***a*. FIGS. **14**A and **14**B illustrate an example in which the fifth wire portion **125** and the sixth wire portion **126** may be spaced apart in the second direction (Y-direction). However, this is merely an example, and the fifth wire portion **125** and the sixth wire portion **126** may be spaced apart from each other in the first direction (X-direction).
(122) However, since the image sensor **111** has a rectangular shape, the fifth wire portion **125** and the sixth wire portion **126** may have a length along the short side of the image sensor **111**, and may be spaced apart from each other along the long side of the image sensor **111**.
(123) FIG. **15** is a diagram illustrating a configuration of a position sensing portion. FIG. **16** is a diagram illustrating a sensing yoke portion and a sensing coil of a position sensing portion, in accordance with one or more embodiments.
(124) Referring to FIGS. **1**, **15** and **16**, the sensor actuator **100** according to an example embodiment may further include a position sensing portion **160**.
(125) For example, when the image sensor **111** moves in the first direction (X-direction), the position of the image sensor **111** may be detected by the position detecting portion **160**, and when the image sensor **111** moves in the second direction (Y-direction), the position of the image sensor **111** may be detected by the position detecting portion **160**.
(126) The position sensing portion **160** may include a first position sensor **170** and a second position sensor **180**. The first position sensor **170** may be used to detect a position of the image sensor **111** in a first direction (X-direction), and the second position sensor **180** may be used to detect a position of the image sensor **111** in a second direction (Y-direction) of the image sensor **111**.
(127) The first position sensor **170** may include a first sensing coil **172** and a first sensing yoke portion **171** (**171***a*, **171***b*). One of the first sensing coil **172** and the first sensing yoke portion **171** may be disposed on the movable body **110**, and the other may be disposed on the fixed body **130**. In an example embodiment, the first sensing yoke portion **171** may be disposed on the housing **131**, and the first sensing coil **172** may be disposed on the sensor substrate **112**. Accordingly, the first sensing coil **172** may be a moving member that moves together with the image sensor **111**.
(128) The first sensing coil **172** and the first sensing yoke portion **171** may be disposed to oppose each other in the optical axis (Z-axis) direction.
(129) The first sensing yoke portion **171** may include a first sensing yoke **171***a* and a second sensing yoke **171***b* spaced apart from each other.
(130) The second position sensor **180** may include a second sensing coil **182** and a second sensing yoke portion **181** (**181***a*, **181***b*). One of the second sensing coil **182** and the second sensing yoke portion **181** may be disposed on the movable body **110**, and the other may be disposed on the fixed body **130**. In an example embodiment, the second sensing yoke portion **181** may be disposed on the housing **131**, and the second sensing coil **182** may be disposed on the sensor substrate **112**. Accordingly, the second sensing yoke portion **181** may be a moving member that moves together with the image sensor **111**.
(131) The second sensing coil **182** and the second sensing yoke portion **181** may be disposed to oppose each other in the optical axis (Z-axis) direction.
(132) The second sensing yoke portion **181** may include a third sensing yoke **181***a* and a fourth sensing yoke **181***b* spaced apart from each other.
(133) Since the configurations and sensing methods of the first position sensor **170** and the second position sensor **180** are the same, only the first position sensor **170** will be described for

convenience of description.
(134) Inductance of the first sensing coil **172** may change according to a change in the position of the first sensing yoke portion **171** opposing each other.
(135) Specifically, when the relative positions of the first sensing coil **172** and the first sensing yoke portion **171** are changed, a magnitude of an eddy current of the first sensing yoke unit **171** affecting inductance of the first sensing coil **172** may change, strength of the magnetic field may change according to the eddy current, and accordingly, inductance of the first sensing coil **172** may be changed.
(136) The first sensing yoke portion **171** may be a conductor or a magnetic material.
(137) The sensor actuator **100** may determine a displacement of the image sensor **111** from changes in inductance of the first sensing coil **172**. For example, the sensor actuator **100** may additionally include at least one capacitor, and the at least one capacitor and the first sensing coil **172** may form a predetermined oscillation circuit.
(138) For example, at least one capacitor may be provided to correspond to the number of first sensing coils **172**, and a capacitor and a first sensing coil **172** may be configured as a predetermined LC oscillator. Additionally, at least one capacitor and the first sensing coil **172** may be configured as a well-known Colpitts oscillator.
(139) The sensor actuator **100** may determine a displacement of the image sensor **111** from a change in a frequency of an oscillation signal generated by the oscillation circuit. Specifically, when inductance of the first sensing coil **172** forming the oscillation circuit changes, a frequency of the oscillation signal generated by the oscillation circuit may change, such that displacement of the image sensor **111** may be detected based on the changes in frequency.
(140) Referring to FIG. **16**, the first sensing yoke portion **171** may include a first sensing yoke **171***a* and a second sensing yoke **171***b*.
(141) In an example embodiment, the first sensing yoke portion **171** may further include a support member **171***c* on which the first sensing yoke **171***a* and the second sensing yoke **171***b* are disposed. The support member **171***c* may be attached to the housing **131**.
(142) The first sensing yoke **171***a* and the second sensing yoke **171***b* may be attached to the support member **171***c*, or the first sensing yoke **171***a* and the second sensing yoke **171***b* may be integrated with the support member **171***c* by an insert injection method, as only an example.
(143) However, an example embodiment thereof is not limited thereto, and the first sensing yoke portion **171** may not include the support member **171***c*. In this example, the first sensing yoke **171***a* and the second sensing yoke **171***b* may be directly attached to the housing **131**, or the first sensing yoke **171***a* and the second sensing yoke **171***b* may be integrated with the housing **131** by an insert injection method.
(144) The first sensing yoke **171***a* and the second sensing yoke **171***b* may be disposed to be spaced apart from each other in the second direction (Y-direction). Additionally, each sensing yoke may be disposed to oppose a portion of the first sensing coil **172**. For example, the first sensing yoke **171***a* and the second sensing yoke **171***b* may be disposed to oppose the first sensing coil **172** in the optical axis (Z-axis) direction.
(145) A direction of a current flowing in a portion of the first sensing coil **172** opposing the first sensing yoke **171***a* may be different from a direction of a current flowing in a portion of the first sensing coil **172** opposing the second sensing yoke **171***b*. In an example embodiment, a direction of the current flowing in a portion of the first sensing coil **172** opposing the first sensing yoke **171***a* may be opposite to a direction of the current flowing in a portion of the first sensing coil **172** opposing the second sensing yoke **171***b*.
(146) The distance between the first sensing yoke **171***a* and the second sensing yoke **171***b* in the second direction (Y-direction) may be shorter than a distance between both ends of the first sensing coil **172** in the second direction (Y-direction).
(147) Each of the first sensing yoke **171***a* and the second sensing yoke **171***b* may have a width that

varies according to coordinates of the direction (e.g., the X-direction) in which the image sensor **111** moves.
(148) The first sensing yoke **171***a* and the second sensing yoke **171***b* may output magnetic flux due to an eddy current, respectively. A magnitude of the eddy current and a magnitude of the magnetic flux may be dependent on each other.
(149) A magnitude of the eddy current formed in each of the first sensing yoke **171***a* and the second sensing yoke **171***b* may be dependent on a width of a portion in which the first sensing yoke **171***a* and the second sensing yoke **171***b* oppose the first sensing coil **172**.
(150) For example, since the first sensing coil **172** may move in the first direction (X-direction) with respect to the first sensing yoke **171***a* and the second sensing yoke **171***b*, a magnitude of the eddy current formed in each of the first sensing yoke **171***a* and the second sensing yoke **171***b* may be dependent on the relative movement of the first sensing coil **172** in the first direction (X-direction).
(151) Since inductance of the first sensing coil **172** may be a sum of the mutual inductance caused by the magnetic flux and self-inductance of the first sensing coil **172** or a difference therebetween, inductance may vary depending on a magnitude of the magnetic flux caused by the eddy current. The position of the image sensor **111** may be detected based on inductance of the first sensing coil **172**.
(152) Since the changes in size of the eddy currents of the first sensing yoke **171***a* and the second sensing yoke **171***b* according to displacement of movement of the image sensor **111** is linear, the position of the image sensor **111** may be precisely detected.
(153) Each of the first sensing yoke **171***a* and the second sensing yoke **171***b* may have a shape in which a width may repeatedly increase or decrease in the direction in which the image sensor **111** moves (e.g., the X-direction). The width may refer to a width in the second direction (Y-direction).
(154) In an example, the first sensing yoke **171***a* may have a shape in which the width may be reduced, increased, decreased, and increased in the first direction (X-direction). The second sensing yoke **171***b* may have a shape in which a width may be repeatedly decreased, increased, decreased, increased, and reduced in the first direction (X-direction).
(155) Each of the first sensing yoke **171***a* and the second sensing yoke **171***b* may have a shape in which the width increases or decreases in one direction, and positions of the first sensing yoke **171***a* and the second sensing yoke **171***b* in which widths increase or decrease may have different shapes.
(156) Each of the first sensing yoke **171***a* and the second sensing yoke **171***b* may have a plurality of minimum widths and a plurality of maximum widths.
(157) A boundary line defining the width of each sensing yoke may have a sinusoidal wave shape.
(158) A winding thickness of the first sensing coil **172** may be greater than a minimum width of each sensing yoke, and may be smaller than a maximum width of each sensing yoke.
(159) A position in which the first sensing yoke **171***a* has a minimum width may be different from a position in which the second sensing yoke **171***b* has a minimum width. Additionally, the position in which the first sensing yoke **171***a* has a maximum width may be different from the position in which the second sensing yoke **171***b* has a maximum width.
(160) Accordingly, coordinates of the image sensor **111** in one direction (e.g., X-direction), corresponding to a maximum width (a maximum width in the second direction (Y-direction)) of the first sensing yoke **171***a*, and coordinates of the image sensor **111** in one direction (e.g., X-direction), corresponding to a maximum width (a maximum width in the second direction (Y-direction)) may be different.
(161) For example, coordinates of the image sensor **111** in the X-direction, corresponding to a minimum width W**1** of the first sensing yoke **171***a*, and coordinates of the image sensor **111** in the X-direction, corresponding to the minimum width of the second sensing yoke **171***b*, may be different from each other, and coordinates in the X-direction, corresponding to the maximum width

W**2** of the first sensing yoke **171***a*, and coordinates in the X-direction, corresponding to the maximum width of the second sensing yoke **171***b*, may be different from each other.
(162) Accordingly, the effect of displacement of the first sensing yoke **171***a* in one direction in a pattern of change in a magnitude of the eddy current of the first sensing yoke **171***a* according to a relative movement of the first sensing coil **172** and the effect of displacement of the second sensing yoke **171***b* in one direction in a pattern of change in a magnitude of the eddy current of the second sensing yoke **171***b* according to a relative movement of the first sensing coil **172** may be complementary to each other.
(163) Accordingly, inductance of the first sensing coil **172** may change stably according to integration of an inductance change factor according to changes in the magnitude of the eddy current of the first sensing yoke **171***a* and an inductance change factor according to changes in the magnitude of the eddy current of the second sensing yoke **171***b*, and the sensor actuator **100** according to an example embodiment may detect movement of the image sensor **111** stably and/or accurately, and also linearly and/or efficiently.
(164) The length of the first sensing yoke **171***a* in the first direction (X-direction) may be one or more periods of a period of a width of the first sensing yoke **171***a*, and the length of the second sensing yoke **171***b* in the first direction (X-direction) may be one or more periods of a period of the second sensing yoke **171***b*.
(165) A width of each of the first sensing yoke **171***a* and the second sensing yoke **171***b* may be repeated every one period. A length of the period of the width of each of the first sensing yoke **171***a* and the second sensing yoke **171***b* in the first direction (X-direction) may vary depending on a movement sensing range of the image sensor **111**.
(166) Due to a difference between coordinates of the image sensor **111** in one direction (e.g., X-direction), corresponding to the maximum width of the first sensing yoke **171***a* and coordinates of the image sensor **111** in one direction (e.g., X-direction), corresponding to the maximum width of the second sensing yoke **171***b*, an output value of the first sensing coil **172** according to the movement of each sensing yoke may be a sine wave having a phase difference of 90 degrees.
(167) Accordingly, an output value obtained by arctangent-processing the output of a sine wave having a phase difference of 90 degrees may be linear with respect to movement of the image sensor **111**.
(168) Each of the first sensing yoke **171***a* and the second sensing yoke **171***b* may include at least one of copper, silver, gold, and aluminum, as only examples. Since copper, silver, gold, and aluminum have relatively high conductivity, an overall magnitude of the eddy current formed in the first sensing yoke **171***a* and the second sensing yoke **171***b* according to a magnetic flux of the first sensing coil **172** may increase, and sensitivity of movement sensing of the image sensor **111** may be further improved.
(169) In example embodiments, the first sensing coil **172** may include a plurality of sensing coils to which an inductance change factor according to changes in the size of an eddy current of the first sensing yoke **171***a* and an inductance change factor according to changes in the size of an eddy current of the second sensing yoke **171***b* are applied, respectively. In this example, the first sensing yoke **171***a* and the second sensing yoke **171***b* may be disposed to oppose different sensing coils.
(170) Since inductance of each of the plurality of sensing coils is used together to generate information on movement of the image sensor **111**, the inductance change factor according to changes in the size of the eddy current of the first sensing yoke **171***a* and the inductance change factor according to changes in in the size of the eddy current of the second sensing yoke **171***b* may be used integrally, and the sensor actuator **100** in an example embodiment may linearly sense movement of the image sensor **111**.
(171) FIGS. **17**A and **17**B are diagrams illustrating changes in a positional relationship between a first sensing yoke portion and a first sensing coil, in accordance with one or more embodiments.
(172) Referring to FIGS. **17**A and **17**B, since a width of the first sensing yoke portion **171** (**171***a*,

171*b*) may change in the moving direction of the image sensor **111**, a change may occur in a region in which the first sensing yoke unit **171** and the first sensing coil **172** overlap in the optical axis (Z-axis) direction as the image sensor **111** moves.
(173) A width of the portions of first sensing yoke **171***a* and the second sensing yoke **171***b* overlapping the first sensing coil **172** in the optical axis (Z-axis) direction may vary according to movement of the sensing yoke **171***b* in the first direction (X-direction). Accordingly, inductance of the first sensing coil **172** may vary according to movement of the image sensor **111** in the first direction (X-direction), and movement of the image sensor **111** in the first direction (X-direction) may be sensed.
(174) FIG. **18** is a graph illustrating inductance of a first sensing coil according to movement of an image sensor in one direction.
(175) Referring to FIG. **18**, a period of a width of the first sensing yoke **171***a* may correspond to a phase of 360 degrees.
(176) When a specific region (e.g., a center of the first sensing coil **172**) of the first sensing coil **172** and a minimum width of the first sensing yoke **171***a* overlap, normalized inductance of the first sensing coil **172** may have a maximum value.
(177) When a specific region (e.g., the center of the first sensing coil **172**) of the first sensing coil **172** and a maximum width of the first sensing yoke **171***a* overlap, normalized inductance of the first sensing coil **172** may have a minimum value.
(178) In an example, the normalized inductance may be a value to which a specific weight is applied to the inductance.
(179) FIG. **19**A is a graph illustrating a plurality of levels of inductance of a first sensing coil corresponding to a first sensing yoke and a second sensing yoke of a sensor actuator, respectively, in accordance with one or more embodiments.
(180) Referring to FIG. **19**A, a phase difference between first inductance L**1** of the first sensing coil **172** corresponding to the first sensing yoke **171***a* and second inductances L**2** of the first sensing coil **172** corresponding to the second sensing yoke **171***b* may be 90 degrees. In an example, the inductance may be a value that is obtained by subtracting a specific value such that an average value may become 0 from the normalized inductance.
(181) FIG. **19**B is a graph illustrating arctangent-processing values of a plurality of levels of inductance illustrated in FIG. **19**A.
(182) Referring to FIG. **19**B, an arctangent-processing value may change linearly with a change in phase.
(183) When the first inductance L**1** and the second inductance L**2** have a phase difference of 90 degrees from each other, one of the first inductance L**1** and the second inductance L**2** may correspond to {sin(phase)} and the other may correspond to {cos(phase)}.
(184) In a trigonometric model, an angle from an origin to one point of the circle may correspond to a phase of one period of the sensing yoke, a distance from the origin to one point of the circle may be r, and an X-direction vector value and a Y-direction vector value from an origin to one point of the circle may be X and Y, respectively.
(185) {sin(phase)} may be (y/r) and {cos(phase)} may be (x/r). {tan(phase)} may be (y/x), {sin(phase)}/{cos(phase)} may be satisfied, and (second inductance)/(first inductance) may be satisfied.
(186) Accordingly, arctan{(second inductance)/(first inductance)} may correspond to a phase of one period of the displacement identification layer, and may be an arctan processing value.
(187) FIG. **20** is a cross-sectional diagram illustrating a camera module **10**, in accordance with one or more embodiments.
(188) Referring to FIG. **20**, the camera module **10** in an example embodiment may include a lens module **200**, a housing **300** and a sensor actuator **100**.
(189) At least one lens that images a subject may be accommodated in the lens module **200**. When

a plurality of lenses are disposed in the lens module **200**, the plurality of lenses may be disposed in the lens module **200** in the optical axis (Z-axis).
(190) In an example, the lens module **200** may have a hollow cylindrical shape.
(191) In another example embodiment, the lens module **200** may include a lens barrel and a lens holder. In this example, at least one lens may be accommodated in the lens barrel, and the lens barrel may be coupled to the lens holder.
(192) The lens module **200** may be accommodated in the housing **300**. Additionally, the housing **300** may be coupled to the housing **131** of the sensor actuator **100**.
(193) The sensor actuator **100** may be the sensor actuator **100** described in the aforementioned example embodiment.
(194) An image sensor **111** may be disposed in the sensor actuator **100**, and the image sensor **111** may move in the first direction (X-direction) and the second direction (Y-direction) based on an operation of the driver **120**, and may rotate about the optical axis (Z-axis) as a rotation axis.
(195) Accordingly, an optical image stabilization function may be performed based on a movement of the image sensor **111**.
(196) The camera module in an example embodiment may perform optical image stabilization by moving the image sensor **111** rather than moving the lens module **200**. Since the image sensor **111** having a relatively light weight moves, the image sensor **111** may move with smaller driving force. Accordingly, the camera module may have a reduced size.
(197) The lens module **200** may move in the optical axis (Z-axis) direction with respect to the housing **300**. Accordingly, a focus may be adjusted by moving the lens module **200** in the optical axis (Z-axis) direction.
(198) The focus adjustment driver may include a magnet **210** and a coil **230** that create a driving force in the optical axis (Z-axis) direction, the magnet **210** may be attached to the lens module **200**, and the coil **230** may be mounted on the housing **300** to oppose the magnet **210**. A substrate for applying power to the coil **230** may be disposed in the housing **300**. The coil **230** may be disposed on one surface of the substrate.
(199) When power is applied to the coil **230**, the lens module **200** may move in the optical axis (Z-axis) direction by electromagnetic force between the magnet **210** and the coil **230**.
(200) When the lens module **200** moves, the ball member B may be disposed between the lens module **200** and the housing **300** to reduce friction between the lens module **200** and the housing **300**. The ball member B may include a plurality of balls.
(201) A guide groove portion for accommodating the ball member B may be formed on at least one of the surfaces of the lens module **200** and the housing **300** opposing each other in a direction perpendicular to the optical axis (Z-axis).
(202) The ball member B may be accommodated in the guide groove portion and may be inserted to a region between the lens module **200** and the housing **300**.
(203) The yoke may be disposed to oppose the magnet **210** in a direction perpendicular to the optical axis (Z-axis). For example, the yoke may be disposed on the other surface of the substrate or the other surface of the coil **230**. Accordingly, the yoke may be disposed to oppose the magnet **210** with the coil **230** interposed therebetween.
(204) An attractive force may act in a direction perpendicular to the optical axis (Z-axis) between the yoke and the magnet **210**.
(205) Accordingly, the ball member B may maintain a contact state with the lens module **200** and the housing **300** by an attractive force between the yoke and the magnet **210**.
(206) A position sensor that opposes the magnet **210** may be disposed on the substrate.
(207) In FIG. **20**, the ball member B may be disposed on the opposite side of the magnet **210**. However, an example embodiment of the ball member B is not limited thereto, and the ball member B may be disposed in a contact state with the lens module **200**, and the housing **300** may be maintained by an attractive force between the magnet **210** and the yoke.

(208) FIG. **21** is a cross-sectional diagram illustrating a camera module **20**, in accordance with one or more embodiments.
(209) Referring to FIG. **21**, the camera module **20**, in an example, may include a housing **300**, a reflective module R, a lens module **200** and a sensor actuator **100**.
(210) In the example embodiment, the optical axis (Z-axis) of the lens module **200** may be directed to a direction perpendicular to the thickness direction (a direction from a front surface to a rear surface of the portable electronic device or vice versa) of the portable electronic device.
(211) In an example, the optical axis (Z-axis) of the lens module **200** may be formed in the width direction or the length direction of the portable electronic device.
(212) When the components included in the camera module are stacked in the thickness direction of the portable electronic device, the thickness of the portable electronic device may increase.
(213) However, in the camera module **20** in the example embodiment, since the optical axis (Z-axis) of the lens module **200** is formed in the width direction or the length direction of the portable electronic device, the thickness of the portable electronic device may be reduced.
(214) In the housing **300**, the reflective module R and the lens module **200** may be disposed. Alternatively, the reflective module R and the lens module **200** may be disposed in different housings, and the housings may be combined with each other.
(215) The reflective module R may be configured to change a traveling direction of light. For example, a direction of light incident into the housing **300** may change to be directed toward the lens module **200** through the reflective module R. The reflective module R may be a mirror or a prism that reflects light.
(216) The sensor actuator **100** may be coupled to the housing **300**.
(217) The sensor actuator **100** may be the sensor actuator **100** described in the aforementioned example embodiment.
(218) An image sensor **111** may be disposed in the sensor actuator **100**, and the image sensor **111** may move in a first direction (X-direction) and a second direction (Y-direction), or may rotate about an optical axis (Z-axis) as a rotation axis.
(219) Accordingly, an optical image stabilization function may be performed based on a movement of the image sensor **111**.
(220) The lens module **200** may move in the optical axis (Z-axis) direction with respect to the housing **300**. Accordingly, a focus may be adjusted by moving the lens module **200** in the optical axis (Z-axis) direction.
(221) Since the configuration of the driver for focus adjustment is the same as the configuration of the focusing driver described with reference to FIG. **20**, a detailed description thereof will not be provided.
(222) According to the aforementioned example embodiments, a sensor actuator and a camera module including the same may improve optical image stabilization performance.
(223) While this disclosure includes specific examples, it will be apparent to one of ordinary skill in the art, after an understanding of the disclosure of this application, that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

Claims

1. A sensor actuator, comprising: a movable body on which an image sensor having an imaging plane is disposed; a fixed body, configured to accommodate the movable body; and a driver, configured to provide a driving force to move the image sensor, wherein the driver comprises a plurality of wires which have lengths that vary to provide the driving force when power is applied to each of the plurality of wires, wherein each of the plurality of wires is configured to have a first end coupled to the fixed body and a second end coupled to the movable body, wherein one of the first end and the second end of each of the plurality of wires is connected to the fixed body or the movable body through an elastic portion, wherein the plurality of wires comprise a first wire portion, a second wire portion, a third wire portion, and a fourth wire portion, wherein each of the first wire portion to the fourth wire portion comprise two wires, and wherein each of the first wire portion to the fourth wire portion is configured to have the first end fixed to the fixed body and the second end fixed to the movable body.
2. The sensor actuator of claim 1, wherein the fixed body comprises a first support plate, and the movable body comprises a second support plate, and wherein the elastic portion is disposed on one of the first support plate and the second support plate.
3. The sensor actuator of claim 1, wherein each of the first wire portion and the second wire portion is configured to provide a driving force to move the movable body in a first direction parallel to the imaging plane, and wherein a moving direction of the movable body based on an operation of the first wire portion and a moving direction of the movable body based on an operation of the second wire portion are opposite to each other.
4. The sensor actuator of claim 3, wherein a length of one of the first wire portion and the second wire portion is reduced when power is applied to the one of the first wire portion and the second wire portion, and wherein the elastic portion connected to wire portions other than the one of the first wire portion and the second wire portion is elastically deformed.
5. The sensor actuator of claim 3, wherein each of the third wire portion and the fourth wire portion is configured to provide a driving force to move the movable body in a second direction parallel to the imaging plane, and wherein a moving direction of the movable body based on an operation of the third wire portion and a moving direction of the movable body based on an operation of the fourth wire portion are opposite to each other.
6. The sensor actuator of claim 5, wherein a length of one of the third wire portion and the fourth wire portion is reduced when power is applied to the one of the third wire portion and the fourth wire portion, and wherein the elastic portion connected to wire portions other than the one of the third wire portion and the fourth wire portion is elastically deformed.
7. The sensor actuator of claim 5, wherein the movable body is configured to rotate by at least two wires which are configured to generate driving forces in opposite directions.
8. The sensor actuator of claim 1, wherein each of the first wire portion and the second wire portion is configured to provide a driving force to move the movable body in a first direction parallel to the imaging plane, wherein the two wires of the first wire portion and the two wires of the second wire portion are spaced apart from each other in a second direction parallel to the imaging plane, and wherein the first direction and the second direction are perpendicular to each other.
9. The sensor actuator of claim 8, wherein each of the third wire portion and the fourth wire portion are configured to provide a driving force to move the movable body in the second direction, respectively, and wherein the two wires of the third wire portion and the two wires of the fourth wire portion are spaced apart from each other in the first direction.
10. The sensor actuator of claim 1, wherein a ball member, configured to support movement of the movable body, is disposed between the movable body and the fixed body.
11. The sensor actuator of claim 1, further comprising: a support substrate configured to support the

movable body such that the movable body moves, wherein the support substrate comprises a deformable portion that is elastically deformed as the movable body moves.

12. The sensor actuator of claim 11, wherein the support substrate further comprises a movable portion on which the movable body is disposed, and a fixed portion coupled to the fixed body, and wherein the elastic portion elastically connects the movable portion to the fixed portion.

13. The sensor actuator of claim 1, further comprising: a position sensing portion configured to sense a position of the image sensor, and comprising a sensing coil disposed on one of the movable body and the fixed body, and a sensing yoke portion disposed on the other of the movable body and the fixed body, wherein the sensing yoke portion comprises a plurality of sensing yokes spaced apart from each other in a direction parallel to the imaging plane, and wherein each sensing yoke is configured to have a width that changes in a moving direction of the image sensor.

14. The sensor actuator of claim 13, wherein the plurality of sensing yokes comprise a first sensing yoke and a second sensing yoke, wherein each of the first sensing yoke and the second sensing yoke opposes the sensing coil in a direction perpendicular to the imaging plane, and wherein each of the first sensing yoke and the second sensing yoke has a width that increases and decreases in a moving direction of the image sensor, and positions of the first sensing yoke and the second sensing yoke in which widths increase or decrease have different shapes.

15. A camera module, comprising: a lens module comprising at least one lens; a housing configured to accommodate the lens module; a first driver configured to move the lens module in an optical axis direction; a fixed body coupled to the housing; a movable body accommodated in the fixed body and comprising an image sensor disposed therein; a second driver configured to provide a driving force to move the image sensor in a first direction and a second direction perpendicular to the optical axis direction, wherein the second driver comprises a plurality of wires which have lengths that vary to provide the driving force when power is applied to each of the plurality of wires, wherein each of the plurality of wires is configured to have a first end coupled to the fixed body and a second end coupled to the movable body, wherein one of the first end and the second end of each wire of the plurality of wires is connected to the fixed body or the movable body through an elastic portion, wherein the plurality of wires comprise a first wire portion, a second wire portion, a third wire portion, and a fourth wire portion, wherein each of the first wire portion to the fourth wire portion comprise two wires, and wherein each of the first wire portion to the fourth wire portion is configured to have the first end fixed to the fixed body and the second end fixed to the movable body.

16. An apparatus, comprising: a camera module, comprising: a movable body on which an image sensor is disposed; a fixed body configured to accommodate the movable body; a driver, configured to provide a driving force to move the image sensor, the driver comprising: wire portions comprising first ends coupled to the fixed body and second ends coupled to the movable body, and configured to provide a driving force to move the moveable body in first and second directions parallel to an imaging plane of the image sensor, and configured to rotate the movable body about an optical axis; wherein each of the wire portions has a length that varies to provide the driving force when power is applied to each of the wire portions, wherein the wire portions comprise a first wire portion, a second wire portion, a third wire portion, and a fourth wire portion, wherein each of the first wire portion to the fourth wire portion comprise two wires, and wherein each of the first wire portion to the fourth wire portion is configured to have the first end fixed to the fixed body and the second end fixed to the movable body.

17. The apparatus of claim 16, wherein each of the wire portions is a shape memory alloy.

18. The apparatus of claim 16, wherein the first direction comprises directions that are opposite to each other, the second direction comprises directions that are opposite to each other, and the first direction is perpendicular to the second direction.